The Prestige: Benchmarking Cognitive Visual Reasoning using Magic Tricks

Shuo Wen[1,2*] **Beixi Du**[3,♠] **Edwin Meriaux**[1,2,4,♠] **Junming Shi**[1,2,♠] **Chloe Si**[1,♠]
David Meger[1,2] **Doina Precup**[1,2] **Gregory Dudek**[1,2]

[1]McGill University [2]Mila – Quebec AI Institute [3]Cornell University
[4]Université Paris-Saclay, CentraleSupélec
♠ Equal contribution, listed alphabetically by last name.

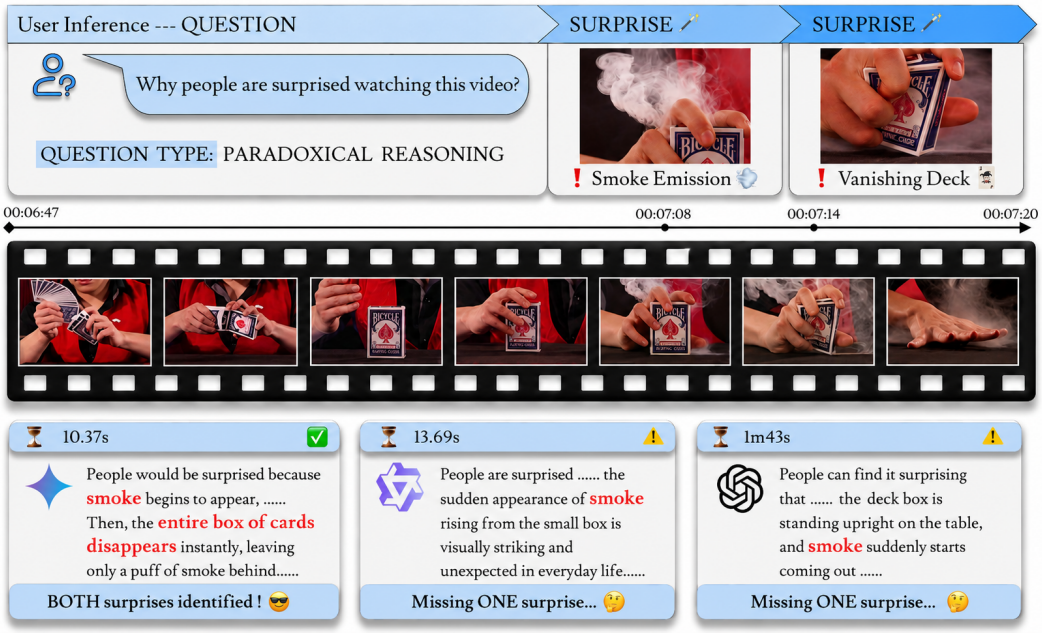


Figure 1: An example prompt to challenge representative VLMs to explain the elements of surprise in a video sequence. Results highlight current limitations in VLM visual reasoning, including partial event detection and false identification. All models are evaluated using their latest web versions at the time of testing. Responses are excerpted, and images are cropped for brevity. Project page

Abstract

We introduce The Prestige, a video understanding benchmark that probes cognitive visual reasoning through professional magic performances, where deliberate misdirection challenges models to track events and explain unexpected outcomes. The benchmark comprises open-ended questions ranging from temporal and causal understanding to robustness against adversarial questions. Evaluation of leading open-weight and proprietary VLMs reveals a substantial gap between even the strongest model and human performance. We also find systematic discrepancies between human and automated grading and identify recurring model failure modes. Ablations further show that models can select correct multiple-choice answers even when they fail to generate correct explanations independently,

*Correspondence to: swen@utexas.edu

40th Conference on Neural Information Processing Systems (NeurIPS 2026). Track on Evaluations and Datasets.

motivating further investigations into question formats. Together, these results highlight limitations in both VLM reasoning and its evaluation, informing the development of more reliable video understanding models and benchmarks.

1 Introduction

The illusion of magic fundamentally exploits the mechanisms of human common-sense reasoning. When a magician pulls a dove from a seemingly empty hat, our surprise stems not from a lack of observation but from our active, unconscious anticipation of logical outcomes. Experiencing this deception relies on what we define as *cognitive flow*: a compositional process that involves tracking long-context visual cues, focusing attention on relevant details, and constructing predictive mental models without explicit language. Since magic deliberately subverts these deeply ingrained, multi-step reasoning processes, it presents an exceptionally rigorous domain for evaluating how intelligent systems parse visual narratives, model expectations, and react to counter-intuitive realities.

Consequently, professional magic performances offer unique, structural advantages for benchmarking the video understanding capabilities of Vision-Language Models (VLMs). First, many of the magic performances are engineered to be comprehended through observation alone, allowing benchmarks to isolate pure visual reasoning without reliance on language guidance or captions. Second, these performances naturally demand long-horizon memory and causal state tracking, where magicians frequently foreshadow elements that are revealed much later, deliberately weaving intricate subplots to misdirect the attentional focus of the audience. Finally, because the core mechanics of advanced sleights are closely guarded secrets within the magic community, it is highly unlikely that models can rely on memorized training data to decipher the tricks.

The rapid development of VLMs [2, 3, 6, 12, 17, 25, 44] has pushed video understanding toward stronger temporal modeling and longer-context reasoning. This trajectory motivates benchmarks to go beyond generic video comprehension to evaluate visual reasoning over temporally distant dependencies, multi-step reasoning, and causal visual clues. Although recent benchmarks have made important progress [4, 8, 15, 31, 34], many still rely on multiple-choice formats [11, 27, 30, 45]. Such formats encourage reasoning shortcuts and only provide limited evidence that models can perform the cognitive visual reasoning required to track hidden state changes, maintain long-horizon expectations, and explain visually surprising outcomes. Our contributions are as follows:

- We introduce *The Prestige*, a novel dataset of professional magic performances designed to evaluate a diverse set of visual reasoning capabilities in VLMs, from short-term perception to complex cognitive-flow reasoning. To comprehensively test these skills, we define a rigorous taxonomy of six distinct question categories, yielding 2,177 open-ended questions grounded in visually sufficient, high-quality magic performances.
- Through extensive evaluations, we demonstrate a substantial performance gap between current state-of-the-art models and human baselines, establishing a rigorous new frontier for evaluating the cognitive visual reasoning abilities of VLMs. Our error analysis further reveals that current VLMs are primarily bottlenecked by limited cognitive flow, weak long-context reference, and unreliable visual identification.
- In addition, we conduct an ablation study demonstrating that multiple-choice formats artificially inflate model performance. This finding strongly advocates for moving toward strictly open-ended question formats to ensure a more rigorous and authentic evaluation of true visual reasoning abilities.

The full dataset can be found at: https://github.com/stevvwen/The_prestige

2 Related Work

2.1 VLM for video understanding

VLMs for videos have rapidly evolved from short-video conversational systems toward stronger temporal modeling and longer-context understanding. Early advancements like Video-LLaVA [17], VideoLLaMA-2 [9], and LLaVA-Video [39] establish strong chat-style interfaces for unified image and video understanding. VideoLLaMA 3 [37] further strengthened this line with a more vision-centric design for both image and video understanding. More recently, models

Dataset	# videos	# QA	Format	Common-sense	Cognitive Flow	Multi-step	Long-context	Annotation
FunQA [31]	4,365	311,950	QA	✓	△	×	×	Manual+LLM
Video-MME-v2 [11]	800	3,200	8-way MC	✓	×	✓	✓	Manual
Causal-VidQA [15]	26,900	107,600	MC	✓	×	×	×	Manual
VRBench [34]	960	8,243	MC&QA	×	×	✓	✓	Manual+LLM
Video-Holmes [8]	270	1,837	MC	✓	×	✓	×	Manual+LLM
CG-Bench [4]	1,219	12,129	MC&QA	×	×	✓	✓	Manual
MMR-V [45]	317	1,257	MC	✓	×	✓	✓	Manual
LongVideoBench [30]	3,763	6,678	MC	×	×	×	✓	Manual
LVBench [27]	103	1,549	MC	×	×	✓	✓	Manual+LLM
Ours	**180**	**2,177**	**Open QA**	✓	✓	✓	✓	Manual

Table 1: Comparison of different video understanding benchmarks. **MC** stands for multiple choice questions, **QA** stands for open Question-Answer questions. **Cognitive Flow** denotes whether the benchmark centrally evaluates reasoning over videos designed to examine the cognitive visual reasoning of the models. △ indicates partial coverage.

such as LongVILA [6], Qwen2.5-VL [3], Qwen3VL [2], InternVL3 [44], GLM-4.6V [14], InternVL3.5 [28], and MiniCPM-V 4.5 [35] have advanced multimodal perception and reasoning, specifically enhancing general-purpose capabilities and long-video comprehension. Further, InternVideo2.5 [29], LongVU [24], and LongVILA-R1 [7] emphasize long-context modeling, efficient compression, and reasoning over extended video streams. In parallel, proprietary systems such as GPT-5 [25] and Gemini 3.1 [12] have further advanced frontier multimodal reasoning capabilities.

2.2 Benchmarks for video understanding

Benchmarks for video understanding have progressed from temporal perception and general QA toward long-context and reasoning-centric evaluation. MVBench [16] and TempCompass [18] evaluate fine-grained temporal understanding, including motion, order, and direction. Video-MME [10] broadens evaluation across video types, durations, and auxiliary modalities, while MLVU [42], LVBench [27], LongVideoBench [30], and ReXTime [5] extend evaluation to longer videos and temporally separated evidence. Video-MME-v2 [11] further introduces a hierarchical evaluation of video reasoning, including counterintuitive comprehension under physical-world reasoning. More recently, CG-Bench evaluates clue-grounded long-video question answering [4], Video-Holmes studies complex reasoning over suspense short films [8], MMR-V focuses on long-range multi-frame evidence and hidden information [45], and VRBench evaluates reasoning in long narrative videos with timestamped intermediate steps [34]. Causal-VidQA [15] studies explanation, prediction, and counterfactual reasoning in video. These settings require models to aggregate evidence over time, but they do not examine the deeper cognitive flow of the model. The closest prior benchmark to our setting is FunQA [31], which includes MagicQA as part of a broader benchmark on counterintuitive videos. However, FunQA mixes magic with humor, creative performances, visual-illusion-style effects, and mainly phrases magic as a local anomaly, whereas our benchmark focuses on staged magic routines in which the performer deliberately controls attention, hides state changes, and delays the reveal. We discuss why we choose magic as the testing bed in Appendix A. The qualitative comparison can be found in Table 1.

3 Methodology

3.1 Video Collection

To rigorously ground the benchmark quality, we perform a selective video collection process where we ensure selected videos meet the Video Quality Standards defined in Appendix D. We ensure the videos are professional, ethical, and high-resolution. The videos must be entirely comprehensible through visual cues alone, independent of audio or text, and not captioned. Additionally, the performances must demonstrate intentional causal coherence and high contextual density. This process is further explained in Section E.1.

3.2 Question and Answer generation

Following rigorous video selection, we systematically construct Question-Answer pairs to comprehensively evaluate diverse aspects of video understanding. To effectively probe the limits of VLMs, we define a taxonomy of core question categories, each targeting distinct aspects of visual, temporal, and common-sense reasoning in multi-step and long-dependency settings. Below are the six categories of questions generated for this dataset. A more in-depth explanation of each category can

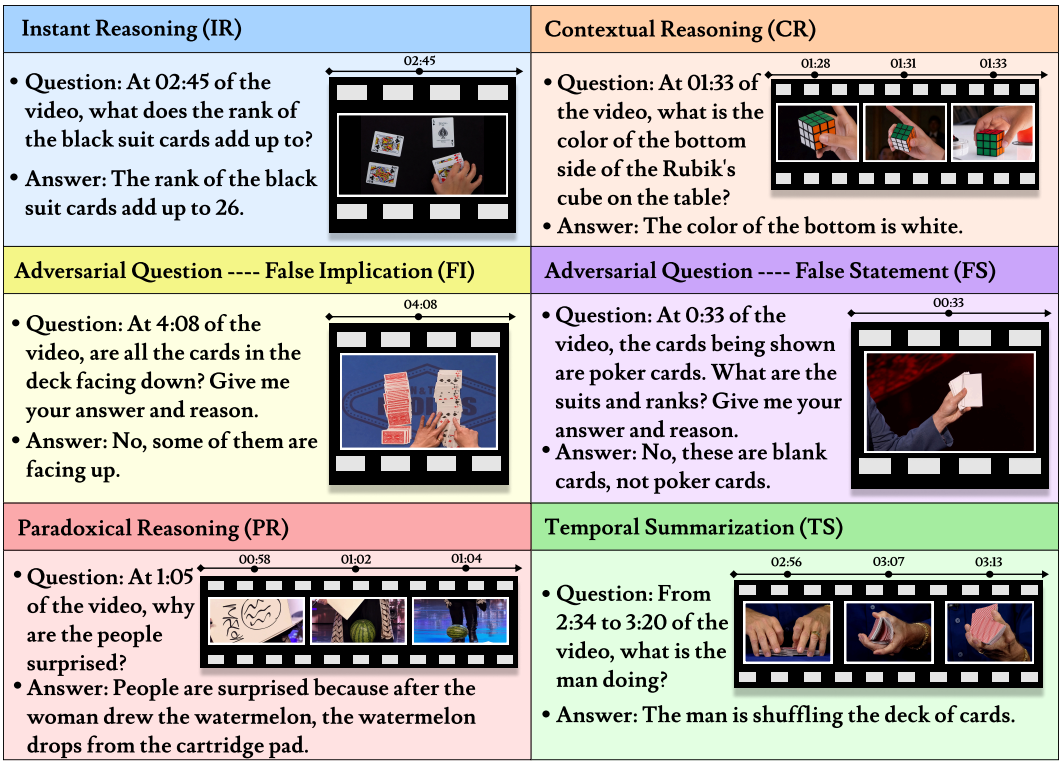


Figure 2: Examples of the six distinct categories of video reasoning tasks evaluated: Instant Reasoning (IR), Contextual Reasoning (CR), Temporal Summarization (TS), Adversarial Questions—False Implication (AQ.FI) and False Statement (AQ.FS), and Paradoxical Reasoning (PR). Each panel displays a sample question, the ground-truth human answer, and a film strip of the relevant video clip. Images are zoomed in for clarity. Representative frames are only selected for demonstration purposes. In actual evaluations, models receive frames sampled uniformly at 1 FPS over the available temporal interval, up to a maximum duration of four minutes.

be found in Section F, and the generation pipeline is described in Section E.2. Visual examples are provided in Figure 2.

Instant Reasoning (IR): These questions aim to assess the instantaneous reasoning abilities of VLMs, based on the visual context of the current frame. Such frame-level analysis forms the necessary foundation for complex temporal understanding and reasoning chains.

Contextual Reasoning (CR): These questions examine the memory and relational reasoning abilities of VLMs. The answers to such questions require context not present in the frame being shown. This context must be understood and constructed by the model from the preceding frames.

Paradoxical Reasoning (PR): These questions are designed to explain the counterintuitive nature of magic tricks. During a magic trick, certain parts of the act are explicitly set up to surprise the audience. Understanding why these scenes are surprising is important as it requires VLMs to perform multi-step reasoning by tracking object states, predicting expected outcomes using common sense, and detecting deviations from observed reality.

Adversarial Questions (AQ): These questions introduce premises, or implications contradicting visual evidence or memory, requiring VLMs to actively reject false information and ground their reasoning in actual observations. **False Implication (AQ.FI)** questions suggest a false proposition through their wording without explicitly asserting it as fact. In contrast, **False Statement (AQ.FS)** questions explicitly make a false claim and construct a follow-up question that assumes the premise is true. A correct response must challenge that premise and provide a correction grounded in visual evidence.

Temporal Summarization (TS): These questions require VLMs to write a compact, high-level summary of events from a given time frame. This does not examine an enumeration of low-level visual details. Models must capture the core actions, avoiding both unnecessary specifics and overly broad generalizations. This category isolates the model's distinct ability to compress long temporal windows into key takeaways.

3.3 Scoring Evaluation

Once these questions are generated, a peer review selection process is conducted to ensure that only questions meeting the Question-Answer Quality Standard (QAQS) are included in the dataset. The QAQS is defined in Section D, and the review process is described in greater detail in Section E.3.

3.4 Statistics of the dataset

The final benchmark contains 180 videos and 2,177 open-ended questions, drawn from unique routines performed by over 150 magicians across multiple platforms. Figure 3 summarizes the distribution of question types. The benchmark covers a broad spectrum of cognitive visual reasoning skills, with question distributions of Contextual Reasoning (36.2%), Paradoxical Reasoning (34.4%), Instant Reasoning (9.4%), False Implication (10.7%), False Statement (4.2%), and Temporal Summarization (5.1%). Beyond question-type diversity, the keyword vocabulary in Figure 4 shows that the questions span varied props and effects, preventing the benchmark from collapsing into a narrow subset of card tricks or object-recognition problems. More comments about the dataset size and question type distribution can be found in Appendix C.

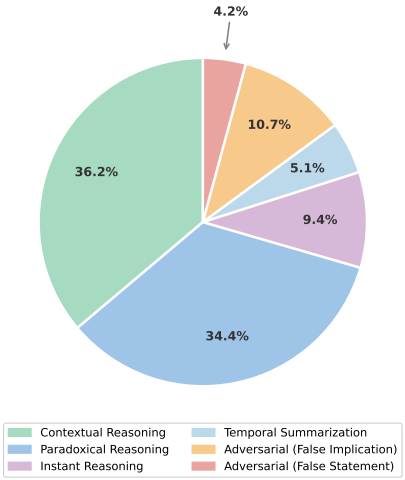


Figure 3: Question Type Distribution

4 Experiments

We evaluate a diverse set of VLMs that natively support long-video or multi-frame image understanding. On the open-source model side, we choose a comprehensive suite of recent representative VLMs: Qwen3VL [2], Qwen3.6 [22], InternVideo2.5 [29], InternVL3.5 [28], MOSS-VL-Instruct [26], Cambrian-S [33], Keye-VL-1.5 [32], MiMo-VL [36], and Kimi-K2.6 [20]. In addition, we include leading proprietary models to establish the current best performance frontiers, including Gemini3.1-Pro [12] and Flash-Lite [13], GPT-5.2 [21], and Qwen3.6-Plus [23]. For consistent comparison under API constraints, all models receive video frames sampled at 1 FPS from a window extending up to four minutes backward from each question's timestamp. To establish the human baseline, we conducted an independent cross-review protocol, where each question was answered by a reviewer other than its author under the same evaluation protocol as the models. Incorrect answers were recorded to compute human pass rates, providing a reference for model performance relative to human visual understanding and reasoning.

We run all open-source models on cluster A100 GPUs, excluding the Qwen family and Kimi-K2.6, which we accessed through the official API portal. For all closed-source models, we utilized their officially provided APIs.



Figure 4: Word cloud

4.1 Benchmark Results

Human baseline Interpretation. The human baseline achieves an overall accuracy of 92.79%, indicating that our dataset is challenging but reliably answerable by careful human viewers. Human performance remains consistently high across question types, with the highest score on PR (94.39%), followed by IR (93.17%) and CR (93.01%). The peak PR score supports our dataset design: humans can reliably solve the central cognitive-flow challenge of the benchmark by tracking the setup, forming an

Model	Size	IR	CR	PR	AQ.FI	AQ.FS	TS	Overall
Open-source models								
Qwen3VL-Thinking [2]	8B	40.00	34.69	9.76	24.46	15.22	17.86	23.84
Qwen3VL-Instruct [2]	8B	46.34	39.90	17.25	18.88	20.65	29.46	29.12
InternVideo2.5-Chat [29]	8B	30.24	27.57	3.74	3.86	5.43	8.93	15.20
InternVL3.5 [28]	8B	25.37	26.81	4.14	19.31	9.78	12.50	16.63
MiMo-VL-SFT-2508 [36]	7B	22.44	22.11	1.87	21.89	15.22	8.04	14.15
MOSS-VL-Instruct [26]	11B	36.59	33.16	12.43	30.47	7.61	28.57	24.76
Qwen3.6 [22]	35A3B	51.71	48.16	28.34	31.76	21.74	37.50	38.26
Cambrian-S [33]	7B	24.88	22.36	4.55	20.17	8.70	10.71	15.07
Keye-VL-1.5 [32]	8B	30.73	27.83	5.61	23.18	14.13	10.71	18.51
Kimi-K2.6 [20]	1.1TA32B	54.15	47.14	31.28	30.47	27.17	40.18	39.37
Proprietary models								
GPT-5.2 [21]	UKN	53.17	46.12	29.41	30.47	15.22	38.39	37.67
Gemini3.1-Flash-Lite [13]	UKN	55.12	49.94	34.89	41.63	18.48	41.96	42.63
Qwen3.6-Plus [23]	UKN	49.76	51.46	35.16	28.33	16.30	39.29	41.11
Gemini3.1-Pro [12]	UKN	52.68	50.44	43.32	45.06	55.43	35.71	47.08
Human baseline	NA	93.17	93.01	94.39	91.42	83.70	90.18	92.79

Table 2: Model accuracy by question type across 14 models over the full dataset. All model results are scored by human evaluation.

expected trajectory, and recognizing how the reveal contradicts that expectation. This indicates that PR questions are visually grounded rather than subjective "surprise" judgments.

The lowest score appears on AQ.FS (83.70%), suggesting that adversarial questions with explicit false premises are difficult even for humans. AQ.FI (91.42%) is higher, indicating that humans are more robust to misleading assumptions compared to false statements. TS also shows a slightly lower score (90.18%), reflecting the variability of open-ended summaries and the difficulty of compressing a temporal window into a concise high-level description without losing key details. Overall, the human baseline highlights both the reliability and difficulty of our benchmark, with the hardest cases involving the same cognitive reasoning abilities we aim to test in VLMs.

Model Performance Analysis. Table 2 reports human-graded accuracy for all 14 models. Gemini3.1-Pro performs best overall at 47.08%, but remains 45.71 percentage points below the human baseline (92.79%). Moreover, no model leads every category: Gemini3.1-Pro achieves the highest PR (43.32%), AQ.FI (45.06%), and AQ.FS (55.43%) scores; Gemini3.1-Flash-Lite leads IR (55.12%) and TS (41.96%); and Qwen3.6-Plus leads CR (51.46%). These complementary strengths indicate that current models have not yet developed uniformly strong visual reasoning across the benchmark's perceptual, temporal, contextual, and adversarial demands.

The strongest open-source models, Kimi-K2.6 (39.37% overall) and Qwen3.6 (38.26%), both outperform GPT-5.2 (37.67%) and substantially exceed the smaller open-source models, whose overall scores range from 14.15% to 29.12%. Nevertheless, even the best model remains well below humans in every category, with particularly large deficits on PR and TS. The results therefore show meaningful progress among larger models while preserving substantial headroom in long-horizon cognitive reasoning and temporal understanding.

4.2 Comparison of Human and Machine Grading

For reproducibility, machine grades are determined by majority vote among Qwen3.7-plus-2026-05-26, Gemini-3.5-flash-lite, and GPT-5.6-Luna, accessed through their official APIs with temperature 0.0 and seed 42. Each grader independently evaluates every model response with the corresponding video frames, using the rubric in Figure 10; the majority vote determines the final binary grade for each question–answer pair. Full machine-grading results are reported in Appendix H.

Table 3 compares human and machine grading across all 14 models. Entries are human accuracy minus machine accuracy, in percentage points; the final row summarizes the mean and standard deviation across models.

Model	IR	CR	PR	AQ.FI	AQ.FS	TS	Overall
Open-source models							
Qwen3VL-Thinking-8B	7.80	3.81	6.42	3.86	10.87	8.04	5.60
Qwen3VL-Instruct-8B	8.78	5.47	8.96	4.29	9.78	13.39	7.44
InternVideo2.5-Chat-8B	6.83	4.32	2.54	0.86	4.34	7.14	3.72
InternVL3.5-8B	7.81	3.18	2.94	3.86	3.26	10.71	4.00
MiMo-VL-SFT-2508-7B	8.29	2.80	1.07	7.30	8.70	3.58	3.49
MOSS-VL-Instruct-11B	4.39	2.66	5.21	5.58	6.52	16.96	4.92
Qwen3.6-35A3B	6.83	4.96	9.22	-2.57	6.52	11.61	6.20
Cambrian-S-7B	6.83	3.17	3.21	8.15	8.70	6.25	4.46
Keye-VL-1.5-8B	5.85	2.93	2.27	2.58	6.52	2.67	3.08
Kimi-K2.6-1.1TA32B	1.95	1.90	6.95	-3.44	1.08	12.50	3.59
Proprietary models							
GPT-5.2	5.37	3.17	1.74	15.45	6.52	-0.90	4.14
Gemini3.1-Flash-Lite	7.80	3.69	5.61	3.86	1.09	6.25	4.78
Qwen3.6-Plus	5.86	3.43	9.09	-0.85	2.17	4.47	5.14
Gemini3.1-Pro	2.44	2.03	5.62	3.86	-1.09	0.89	3.30
AVG *SD*	6.20 *2.08*	3.39 *1.01*	5.06 *2.83*	3.77 *4.76*	5.36 *3.63*	7.40 *5.13*	4.56 *1.23*

Table 3: Signed percentage-point difference between human and machine grading scores, computed as human score minus machine score. The final row reports the mean for each column, with the corresponding standard deviation.

Human grading yields higher overall scores for every model, with a mean gap of 4.56 percentage points (SD 1.23), ranging from 3.08 for Keye-VL-1.5 to 7.44 for Qwen3VL-Instruct. The gap varies by question type: TS has the largest mean difference (7.40 points), while CR has the smallest (3.39 points). TS and AQ.FI also show the greatest variation across models (SD 5.13 and 4.76, respectively). Negative differences in several AQ.FI entries, Gemini3.1-Pro's AQ.FS score, and GPT-5.2's TS score show that machine grading is not uniformly stricter. These aggregate differences indicate sensitivity to both the model and question type, but do not establish agreement on individual answers.

4.3 Error Analysis

To better understand the qualitative failure modes of frontier VLMs on *The Prestige*, we conduct an error analysis on representative model outputs. Specifically, we randomly sample 100 incorrectly answered questions from each of Gemini3.1-Pro [12], GPT-5.2 [21], and Kimi-K2.6 [20]. Each sampled error is manually assigned to one of seven error categories.

Incorrect Visual Identification (IVI). IVI includes errors caused by incorrect recognition of evidence directly visible at the queried timestamp, such as misidentifying objects, misreading card ranks or suits, or miscounting items.

Context Referral Deficiency (CRD). CRD captures failures in long-context reference and state tracking, where the model fails to retrieve information established earlier, such as a previously shown object identity or a hidden object location. These errors occur when the answer is not directly visible at the queried frame but can be inferred by linking the current scene to prior evidence.

Lack of Cognitive Flow (LCF). LCF captures failures to follow the evolving causal structure of a magic routine. The model may recognize local visual events but fail to connect the setup, misdirection, and reveal, or may reuse an earlier surprise to explain a later, distinct one.

Lack of Robustness (LR). LR describes failures to resist misleading or false premises, especially in adversarial questions. These errors occur when the model accepts an incorrect assumption in the questions instead of rejecting it and grounding the answer in visual evidence.

Incorrect Output Formatting (IOF). IOF includes failures to follow the required response format, such as never-ending responses, refusals to answer, or missing required components.

Human Intention Mismatch (HIM). HIM includes answers that are plausible or partially correct but fail to match the intended specificity or reasoning target of the question. Examples include giving generic explanations such as "sleight of hand," over-detailing temporal summaries, or omitting explicitly requested reasoning.

Hallucination (HAL). HAL refers to unsupported visual or causal claims, including invented objects, nonexistent actions, or incorrect surprises that substantially diverge from the ground-truth.

Figure 5 shows that LCF, CRD, and IVI together account for 67%, 66%, and 65% of sampled errors in Gemini3.1-Pro, GPT-5.2, and Kimi-K2.6, respectively, identifying cognitive-flow reasoning, long-context reference, and local visual identification as the dominant failure modes. HAL and IOF are less frequent, together accounting for 13%, 11%, and 12%, respectively. Gemini3.1-Pro has the fewest sampled CRD errors and the highest CR accuracy among these models, consistent with stronger long-context reference ability.

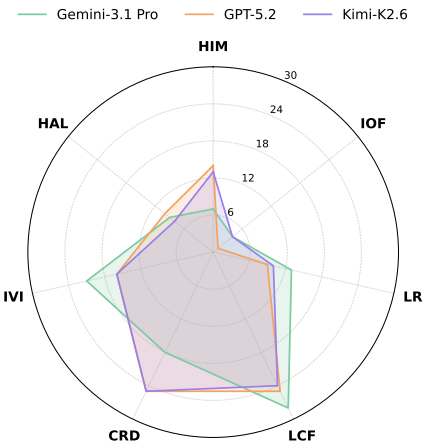


Figure 5: Error Type Distribution

These comparisons are descriptive: each model contributes only 100 randomly sampled errors, and small differences may reflect sampling variation or borderline annotation decisions despite a shared rubric. The sampled question-type distributions are broadly comparable (Figure 6), but associations between question and error types may also affect the observed differences.

Together with Table 2, these findings highlight recurring difficulties in rejecting false premises and following a routine's cognitive flow: identifying the central surprise, linking it to the setup, and summarizing the essential steps.

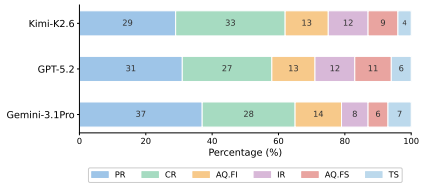


Figure 6: Question Type Distribution of Sampled Errors

5 Ablations on the effects of MC Questions

We evaluate the effect of question format on Qwen3.6-Plus, Kimi-K2.6, and Gemini3.1-Flash-Lite using two samples: 200 questions randomly sampled from the benchmark without conditioning on open-ended correctness, and a separate diagnostic sample of 50 questions from human evaluation that all three models failed to answer in the open-ended setting. For both samples, we compare open-ended answering with 8-way and 4-way multiple-choice answering using the same question-conversion procedure. Following prior work [40, 8], we use an LLM to generate seven distractors for each question, designed to be semantically similar to the correct answer but factually incorrect. The 8-way setting includes the correct answer and all seven distractors; the 4-way setting includes the correct answer and three distractors randomly sampled from that pool. We randomly permute the option order to mitigate option-position bias and extract the selected answer using regular expression matching.

Randomly sampled questions. Figure 7(b) reports the comparison on 200 randomly sampled questions. Qwen3.6-Plus achieves 34.0%, 67.0%, and 71.0% accuracy in the open-ended, 8-way, and 4-way settings, respectively. The corresponding scores are 39.0%, 68.0%, and 74.0% for Gemini3.1-Flash-Lite, and 39.5%, 66.5%, and 69.5% for Kimi-K2.6. Averaged across the three models, accuracy increases from 37.5% for open-ended answering to 67.2% for 8-way and 71.5% for 4-way multiple choice, corresponding to gains of 29.7 and 34.0 percentage points, respectively. Thus, the increase in measured accuracy is also observed on a sample selected independently of open-ended performance.

Failure-selected questions. Figure 7(a) provides a complementary diagnostic on 50 questions that all three models failed in the open-ended setting. The open-ended accuracy is therefore 0% by construction. In the 8-way setting, Qwen3.6-Plus, Kimi-K2.6, and Gemini3.1-Flash-Lite achieve

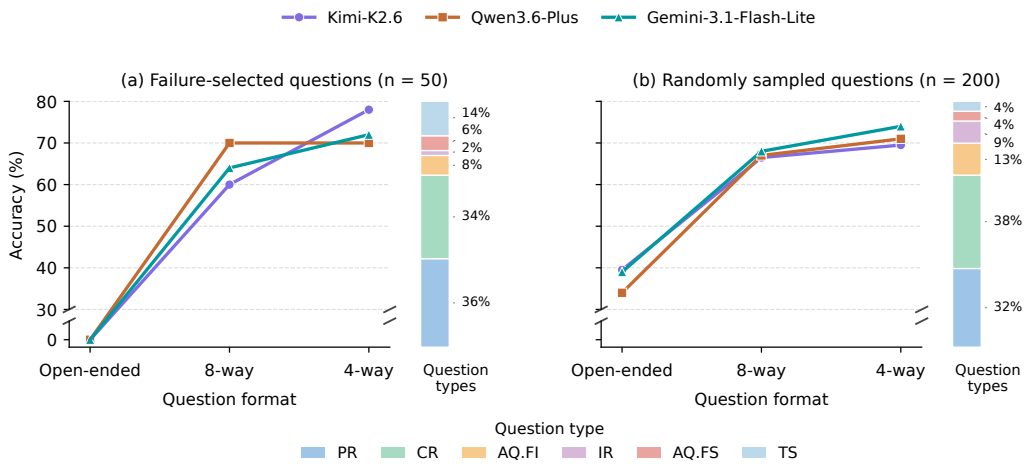


Figure 7: Ablation study on the effects of multiple-choice questions for (a) 50 failure-selected and (b) 200 randomly sampled questions. Stacked bars show the question-type distributions.

70%, 60%, and 64%, respectively; their 4-way scores are 70%, 78%, and 72%. The mean 4-way gain is 73.3 percentage points. This gain is conditional on jointly failed questions and should not be interpreted as the average effect across the benchmark.

Together, these comparisons show that providing candidate answers substantially increases measured accuracy and can conceal failures to generate an adequate answer independently. These findings support open-ended evaluation as a complementary diagnostic and motivate further study of how question format shapes the assessment of model capabilities for tasks requiring qualitative reasoning, rather than the unambiguous numerical answers found in many mathematical problems.

6 Discussion & Conclusion

In this paper, we introduce *The Prestige*, a video understanding benchmark for testing cognitive visual reasoning of VLMs featuring professional magic performances, consisting of 2,177 open-ended questions that test short-term, multistep, and long-range visual reasoning, as well as resistance to adversarial questions. Across 14 leading proprietary and open-source VLMs, the best model, Gemini3.1-Pro, scores 47.08%, leaving a 45.71 percentage-point gap below the 92.79% human baseline. Sampled error evaluation further reveals recurring failures in cognitive flow (LCF), context referral (CRD), and visual identification (IVI). Specifically, we find that VLMs consistently struggle when answering requires recalling an object absent from the current frame. They sometimes miss magic surprises that human reviewers commonly perceive and frequently fail to explicitly reject false information in adversarial settings. These failure modes reveal limited robustness in long-horizon tracking and multi-step reasoning within adversarial narratives.

Our comparison of human and majority-vote machine grading shows that human evaluation yields higher overall scores for all 14 models, with a mean difference of 4.56 percentage points and a standard deviation of 1.23 across models. The direction and magnitude of the differences vary by question category, with machine scores exceeding human scores in some model–category combinations. These aggregate differences indicate sensitivity to the grading protocol but do not establish agreement on individual responses or explain the mechanisms underlying grader disagreement. They motivate further validation of automatic grading for open-ended visual reasoning tasks.

Furthermore, on 200 randomly sampled questions, mean accuracy across three models rises from 37.5% with open-ended answers to 67.2% with eight choices and 71.5% with four choices. The 34.0-percentage-point gain in the 4-way setting shows that question format substantially affects measured performance beyond the failure-selected sample. The larger 73.3-percentage-point gain on 50 jointly failed questions is a conditional diagnostic rather than a benchmark-wide estimate. These comparisons demonstrate that selecting a supplied answer and generating an adequate answer

independently are substantially different evaluation tasks, supporting the diagnostic value of open-ended responses.

Limitation We only release YouTube video identifiers rather than full video files to respect licensing and platform terms, so source videos may become unavailable over time. We have archived local copies, but sharing them for reproducibility depends on creator permission and compliance with applicable licenses.

Impact Looking further, our dataset may also inform training for complex visual scenarios, with potential downstream value for robotics.

Acknowledgement

Shuo Wen and Edwin Meriaux are supported by the FRQNT scholarship (The Fonds de recherche du Québec – Nature et technologies). The research is supported by Fill in funding and computing infrastructure at MILA- Quebec AI Institute. The authors are grateful to the anonymous reviewers and Area Chairs for their thoughtful comments during the review of the paper.

References

[1] Md. Atabuzzaman Ali, Ali Asgarov, and Chris Thomas. Benchmarking and mitigating mcqa selection bias of large vision-language models. *arXiv preprint arXiv:2509.16805*, 2025.

[2] Shuai Bai, Yuxuan Cai, Ruizhe Chen, Keqin Chen, Xionghui Chen, Zesen Cheng, Lianghao Deng, Wei Ding, Chang Gao, Chunjiang Ge, et al. Qwen3-vl technical report. *arXiv preprint arXiv:2511.21631*, 2025.

[3] Shuai Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Sibo Song, Kai Dang, Peng Wang, Shijie Wang, Jun Tang, et al. Qwen2.5-vl technical report. *arXiv preprint arXiv:2502.13923*, 2025.

[4] Guo Chen, Yicheng Liu, Yifei Huang, Yuping He, Baoqi Pei, Jilan Xu, Yali Wang, Tong Lu, and Limin Wang. Cg-bench: Clue-grounded question answering benchmark for long video understanding. *arXiv preprint arXiv:2412.12075*, 2024.

[5] Jr-Jen Chen, Yu-Chien Liao, Hsi-Che Lin, Yu-Chu Yu, Yen-Chun Chen, and Yu-Chiang F Wang. Rextime: A benchmark suite for reasoning-across-time in videos. *Advances in Neural Information Processing Systems*, 37:28662–28673, 2024.

[6] Yukang Chen, Fuzhao Xue, Dacheng Li, Qinghao Hu, Ligeng Zhu, Xiuyu Li, Yunhao Fang, Haotian Tang, Shang Yang, Zhijian Liu, et al. Longvila: Scaling long-context visual language models for long videos. *arXiv preprint arXiv:2408.10188*, 2024.

[7] Yukang Chen, Wei Huang, Baifeng Shi, Qinghao Hu, Hanrong Ye, Ligeng Zhu, Zhijian Liu, Pavlo Molchanov, Jan Kautz, Xiaojuan Qi, et al. Scaling rl to long videos. *arXiv preprint arXiv:2507.07966*, 2025.

[8] Junhao Cheng, Yuying Ge, Teng Wang, Yixiao Ge, Jing Liao, and Ying Shan. Video-holmes: Can mllm think like holmes for complex video reasoning? *arXiv preprint arXiv:2505.21374*, 2025.

[9] Zesen Cheng, Sicong Leng, Hang Zhang, Yifei Xin, Xin Li, Guanzheng Chen, Yongxin Zhu, Wenqi Zhang, Ziyang Luo, Deli Zhao, et al. Videollama 2: Advancing spatial-temporal modeling and audio understanding in video-llms. *arXiv preprint arXiv:2406.07476*, 2024.

[10] Chaoyou Fu, Yuhan Dai, Yongdong Luo, Lei Li, Shuhuai Ren, Renrui Zhang, Zihan Wang, Chenyu Zhou, Yunhang Shen, Mengdan Zhang, et al. Video-mme: The first-ever comprehensive evaluation benchmark of multi-modal llms in video analysis. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 24108–24118, 2025.

[11] Chaoyou Fu, Haozhi Yuan, Yuhao Dong, Yi-Fan Zhang, Yunhang Shen, Xiaoxing Hu, Xueying Li, Jinsen Su, Chengwu Long, Xiaoyao Xie, et al. Video-mme-v2: Towards the next stage in benchmarks for comprehensive video understanding. *arXiv preprint arXiv:2604.05015*, 2026.

[12] Google DeepMind. Gemini 3.1 pro. https://deepmind.google/models/model-cards/gemini-3-1-pro/, 2026.

[13] Google DeepMind. Gemini 3.1 Flash-Lite Model Card. https://deepmind.google/models/model-cards/gemini-3-1-flash-lite/, 2026.

[14] Wenyi Hong, Wenmeng Yu, Xiaotao Gu, Guo Wang, Guobing Gan, Haomiao Tang, Jiale Cheng, Ji Qi, Junhui Ji, Lihang Pan, et al. Glm-4.5 v and glm-4.1 v-thinking: Towards versatile multimodal reasoning with scalable reinforcement learning. *arXiv preprint arXiv:2507.01006*, 2025.

[15] Jiangtong Li, Li Niu, and Liqing Zhang. From representation to reasoning: Towards both evidence and commonsense reasoning for video question-answering. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 21273–21282, 2022.

[16] Kunchang Li, Yali Wang, Yinan He, Yizhuo Li, Yi Wang, Yi Liu, Zun Wang, Jilan Xu, Guo Chen, Ping Luo, et al. Mvbench: A comprehensive multi-modal video understanding benchmark. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 22195–22206, 2024.

[17] Bin Lin, Yang Ye, Bin Zhu, Jiaxi Cui, Munan Ning, Peng Jin, and Li Yuan. Video-llava: Learning united visual representation by alignment before projection. In *Proceedings of the 2024 conference on empirical methods in natural language processing*, pages 5971–5984, 2024.

[18] Yuanxin Liu, Shicheng Li, Yi Liu, Yuxiang Wang, Shuhuai Ren, Lei Li, Sishuo Chen, Xu Sun, and Lu Hou. Tempcompass: Do video llms really understand videos? In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 8731–8772, 2024.

[19] Olga Loginova, Oleksandr Bezrukov, and Alexey Kravets. Addressing blind guessing: Calibration of selection bias in multiple-choice question answering by video language models. *arXiv preprint arXiv:2410.14248*, 2024.

[20] Moonshot AI. Kimi K2.6: Advancing open-source coding. https://www.kimi.com/blog/kimi-k2-6, 2026.

[21] OpenAI. Introducing GPT-5.2. https://openai.com/index/introducing-gpt-5-2/, 2025.

[22] Qwen Team. Qwen3.6-35B-A3B: Agentic coding power, now open to all, April 2026. URL https://qwen.ai/blog?id=qwen3.6-35b-a3b.

[23] Qwen Team. Qwen3.6-Plus: Towards Real World Agents. https://qwen.ai/blog?id=qwen3.6, 2026.

[24] Xiaoqian Shen, Yunyang Xiong, Changsheng Zhao, Lemeng Wu, Jun Chen, Chenchen Zhu, Zechun Liu, Fanyi Xiao, Balakrishnan Varadarajan, Florian Bordes, et al. Longvu: Spatiotemporal adaptive compression for long video-language understanding. *arXiv preprint arXiv:2410.17434*, 2024.

[25] Aaditya Singh, Adam Fry, Adam Perelman, Adam Tart, Adi Ganesh, Ahmed El-Kishky, Aidan McLaughlin, Aiden Low, AJ Ostrow, Akhila Ananthram, et al. Openai gpt-5 system card. *arXiv preprint arXiv:2601.03267*, 2025.

[26] OpenMOSS Team. MOSS-VL Technical Report. https://github.com/OpenMOSS/MOSS-VL, 2026. GitHub repository.

[27] Weihan Wang, Zehai He, Wenyi Hong, Yean Cheng, Xiaohan Zhang, Ji Qi, Ming Ding, Xiaotao Gu, Shiyu Huang, Bin Xu, et al. Lvbench: An extreme long video understanding benchmark. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 22958–22967, 2025.

[28] Weiyun Wang, Zhangwei Gao, Lixin Gu, Hengjun Pu, Long Cui, Xingguang Wei, Zhaoyang Liu, Linglin Jing, Shenglong Ye, Jie Shao, et al. Internvl3. 5: Advancing open-source multimodal models in versatility, reasoning, and efficiency. *arXiv preprint arXiv:2508.18265*, 2025.

[29] Yi Wang, Xinhao Li, Ziang Yan, Yinan He, Jiashuo Yu, Xiangyu Zeng, Chenting Wang, Changlian Ma, Haian Huang, Jianfei Gao, et al. Internvideo2. 5: Empowering video mllms with long and rich context modeling. *arXiv preprint arXiv:2501.12386*, 2025.

[30] Haoning Wu, Dongxu Li, Bei Chen, and Junnan Li. Longvideobench: A benchmark for long-context interleaved video-language understanding. *Advances in Neural Information Processing Systems*, 37:28828–28857, 2024.

[31] Binzhu Xie, Sicheng Zhang, Zitang Zhou, Bo Li, Yuanhan Zhang, Jack Hessel, Jingkang Yang, and Ziwei Liu. Funqa: Towards surprising video comprehension. In *European Conference on Computer Vision*, pages 39–57. Springer, 2024.

[32] Biao Yang, Bin Wen, Boyang Ding, Changyi Liu, Chenglong Chu, Chengru Song, Chongling Rao, Chuan Yi, Da Li, Dunju Zang, et al. Kwai keye-vl 1.5 technical report. *arXiv preprint arXiv:2509.01563*, 2025.

[33] Shusheng Yang, Jihan Yang, Pinzhi Huang, Ellis Brown, Zihao Yang, Yue Yu, Shengbang Tong, Zihan Zheng, Yifan Xu, Muhan Wang, et al. Cambrian-s: Towards spatial supersensing in video. In *International Conference on Learning Representations*, volume 2026, pages 78185–78225, 2026.

[34] Jiashuo Yu, Yue Wu, Meng Chu, Zhifei Ren, Zizheng Huang, Pei Chu, Ruijie Zhang, Yinan He, Qirui Li, Songze Li, et al. Vrbench: A benchmark for multi-step reasoning in long narrative videos. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 21655–21666, 2025.

[35] Tianyu Yu, Zefan Wang, Chongyi Wang, Fuwei Huang, Wenshuo Ma, Zhihui He, Tianchi Cai, Weize Chen, Yuxiang Huang, Yuanqian Zhao, et al. Minicpm-v 4.5: Cooking efficient mllms via architecture, data, and training recipe. *arXiv preprint arXiv:2509.18154*, 2025.

[36] Zihao Yue, Zhenru Lin, Yifan Song, Weikun Wang, Shuhuai Ren, Shuhao Gu, Shicheng Li, Peidian Li, Liang Zhao, Lei Li, et al. Mimo-vl technical report. *arXiv preprint arXiv:2506.03569*, 2025.

[37] Boqiang Zhang, Kehan Li, Zesen Cheng, Zhiqiang Hu, Yuqian Yuan, Guanzheng Chen, Sicong Leng, Yuming Jiang, Hang Zhang, Xin Li, et al. Videollama 3: Frontier multimodal foundation models for image and video understanding. *arXiv preprint arXiv:2501.13106*, 2025.

[38] Hao Zhang, Chen Li, and Basura Fernando. Mitigating easy option bias in multiple-choice question answering. *arXiv preprint arXiv:2508.13428*, 2025.

[39] Yuanhan Zhang, Jinming Wu, Wei Li, Bo Li, Zejun Ma, Ziwei Liu, and Chunyuan Li. Llava-video: Video instruction tuning with synthetic data. *arXiv preprint arXiv:2410.02713*, 2024.

[40] Yuanhan Zhang, Yunice Chew, Yuhao Dong, Aria Leo, Bo Hu, and Ziwei Liu. Towards video thinking test: A holistic benchmark for advanced video reasoning and understanding. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 20626–20636, 2025.

[41] Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric Xing, et al. Judging llm-as-a-judge with mt-bench and chatbot arena. *Advances in neural information processing systems*, 36:46595–46623, 2023.

[42] Junjie Zhou, Yan Shu, Bo Zhao, Boya Wu, Zhengyang Liang, Shitao Xiao, Minghao Qin, Xi Yang, Yongping Xiong, Bo Zhang, et al. Mlvu: Benchmarking multi-task long video understanding. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 13691–13701, 2025.

[43] Xin Zhou, Kisub Kim, Ting Zhang, Martin Weyssow, Luís F Gomes, Guang Yang, Kui Liu, Xin Xia, and David Lo. Se-jury: An llm-as-ensemble-judge metric for narrowing the gap with human evaluation in se. In *2025 40th IEEE/ACM International Conference on Automated Software Engineering (ASE)*, pages 2606–2618. IEEE, 2025.

[44] Jinguo Zhu, Weiyun Wang, Zhe Chen, Zhaoyang Liu, Shenglong Ye, Lixin Gu, Hao Tian, Yuchen Duan, Weijie Su, Jie Shao, et al. Internvl3: Exploring advanced training and test-time recipes for open-source multimodal models. *arXiv preprint arXiv:2504.10479*, 2025.

[45] Kejian Zhu, Zhuoran Jin, Hongbang Yuan, Jiachun Li, Shangqing Tu, Pengfei Cao, Yubo Chen, Kang Liu, and Jun Zhao. Mmr-v: What's left unsaid? a benchmark for multimodal deep reasoning in videos. *arXiv preprint arXiv:2506.04141*, 2025.

A Magic as a testbed for cognitive visual reasoning

Professional magic performances provide a structured testbed for cognitive visual reasoning because their reveals are designed around a discrepancy between an expected outcome and a visually observed one. Explaining why a reveal is surprising requires more than recognizing objects or describing isolated actions. A model must identify the relevant visual evidence, track how the depicted situation changes over time, use commonsense knowledge to infer what would ordinarily occur, and explain how the observed outcome departs from that expectation.

Expectation Formation and Violation Magic performances establish expectations through visible arrangements and actions, such as placing an object inside a container or covering it with a solid barrier. Interpreting the subsequent reveal requires applying commonsense physical and logical expectations, including object permanence, containment, and continuity. The model must then compare the observed outcome with the expected one and identify the specific discrepancy. Paradoxical Reasoning questions in our dataset formalize this process by requiring an explanation that connects the established setup, the ordinarily expected outcome, and the observed departure.

Temporal State Tracking The evidence needed to form an expectation may occur well before the reveal. Intervening actions, occlusions, and misdirection can separate the relevant observations across time. A model must therefore preserve object identities and states, track visible changes, and retrieve the earlier evidence relevant to the final outcome. This temporal integration is necessary to determine not only what occurred, but why the reveal is significant in light of the preceding performance.

Evidence Focus and Visual Grounding Magic also requires distinguishing reasoning-relevant evidence from visually noticeable but irrelevant actions introduced through misdirection. Because our dataset excludes audio narration, models must ground their explanations in the visible setup, actions, and reveal. A successful answer must connect these observations into a coherent logic chain rather than merely describe the final frame or state generically that the outcome is surprising.

The Prestige evaluates behavioral evidence of cognitive visual reasoning: whether a model can integrate visual observations across time, apply commonsense expectations, and explain an observed violation.

B Why moving away from MC

A natural design choice for a video-understanding benchmark is multiple-choice question answering (MCQA), since it offers transparent, deterministic scoring, and trivial aggregation. We deliberately reject this format for *The Prestige*, because a growing body of evidence shows that MCQA systematically inflates the measured performance of vision-language models (VLMs) without reflecting genuine visual understanding. We summarize the failure modes that motivated our open-ended design.

Token and positional selection bias. VLMs exhibit a well-documented tendency to favor particular options and particular positions in the option list, independent of the visual evidence [1, 19]. Prior work, which evaluates Qwen2.5-VL across difficulty-stratified MCQA benchmarks, reports that the model gravitates toward the token `A` on easy items and toward whichever option appears first when the option order is permuted [1].

Shortcutting and uninspectable reasoning. Beyond token bias, MCQA difficulty is largely determined by its distractors, which let models eliminate implausible options using world knowledge alone without ever engaging the video content [4]. Zhang et al. identify a related shortcut across six VQA benchmarks (including NExT-QA, STAR, and Video-MME) and four VLM families: providing only the vision input and the options (withholding the question entirely) recovers a mean accuracy of 51.57%, just 9.54 points below the full "vision+question+options" setting [38]. Even when MCQA scoring is accurate, the format actively suppresses the reasoning trace we wish to evaluate. A correct option **A** reveals nothing about whether the model tracked the deck across a misdirection or fabricated a causal story, and an incorrect **A** reveals nothing about which step of the chain failed.

Why open-ended answers. The same properties that make MCQA brittle make open-QA evaluation a natural fit for our setting. By forcing the model to commit to a specific causal account in its own words, the open-ended format eliminates the token, positional, and elimination shortcuts

described above and exposes fabrication directly. A model that has not tracked the hidden state of the trick will either decline or hallucinate, and either failure is visible to the grader. Crucially, in the context of our AQ (Section 3.2), a free-form answer can surface sycophantic agreement with a false statement, whereas an MCQ over an AQ.FS can provide multiple clues to the assumption we are testing. This consideration has driven recent long-video benchmarks such as CG-Bench [4] and VRBench [34] to shift partially or wholly toward open-ended evaluation, and we adopt the same format throughout the dataset.

C Dataset comments

Our dataset is smaller than general-purpose video-QA benchmarks. However, our goal is to evaluate cognitive visual reasoning through open-ended questions, where we prioritize video quality, question depth, and annotation reliability over dataset size. The concentration in CR and PR is intentional and reflects the benchmark's central objective: magic performances are especially effective for evaluating long-horizon state tracking and cognitive reasoning about outcomes that violate established expectations. The distribution is therefore weighted toward the target capabilities rather than designed to uniformly cover all video-understanding skills.

We evaluate all models using frames sampled at 1 FPS to accommodate model and API constraints. We encourage future users to evaluate the videos at their native frame rates to preserve fine-grained temporal information and make fuller use of the dataset.

D Quality Standards

Across this dataset, high-quality videos and question-answer pairs are fundamental. In this section, we outline our Video Quality Standards (VQS) for source video collection and Question-Answer Quality Standards (QAQS) for dataset construction.

Video Quality Standard:

- **Magic Quality** The video must depict a professional magic performance. Amateur or homemade recordings are excluded, as they tend to exhibit performance flaws, insufficient complexity, and unnatural pacing.
- **Ethical Quality** The selected video must not contain content that is violent, unethical, or otherwise inappropriate. This applies to both the magician and the platforms presenting the act.
- **Temporal length.** The magic performance must exceed 90 seconds in duration, presented in its full, unclipped form. This lower bound ensures the inclusion of multi-stage routines and long-range foreshadowing, thereby stress-testing the model's temporal reasoning, state tracking, and memory. Consequently, VLMs are required to autonomously identify the proper performance window and reason over it without external segmentation.
- **Visual fidelity.** Each video must be recorded at a minimum resolution of 720p, most are 1080p, under adequate lighting conditions. Human annotators verify that every visual cue required to answer the associated questions is clearly discernible. Videos exhibiting abrupt cuts, pronounced camera jitter, or rapid viewpoint switching are excluded.
- **Visual self-sufficiency.** The performance must be fully comprehensible from visual information alone, without reliance on audio narration or dialogue. Videos containing hardcoded captions are excluded, ensuring that evaluation reflects genuine visual understanding rather than incidental textual cues.
- **Minimal Text:** All videos contain minimal on-screen text and no captions to prevent models from over-relying on textual cues instead of visual understanding.
- **Causal coherence.** The video must exhibit a clear causal structure and present commonsense-violating surprises that are intentionally orchestrated by the performer. Surprises arising from technical glitches, editing artifacts, or other factors outside the performer's control are excluded.
- **Contextual density.** Each video must be visually rich and contain multiple distinct magic surprises, ensuring that the associated question–answer pairs remain non-trivial and that the benchmark probes a model's capacity to process dense, long-horizon visual evidence.

Question-Answer Quality Standard:

- **Triviality filter.** Questions answerable from the question text alone, without watching the video, are removed.
- **Insufficient-context filter.** Questions whose visual grounding is impossible (relevant frames off-screen, motion-blurred beyond recognition, occluded by the performer's body, or cut away by the camera) are removed. A careful human viewer with access to the cited timestep must be able to answer from the visual content alone.
- **External-knowledge filter.** Questions requiring outside knowledge (a magician's biography, an unobservable backstage detail, magic-industry trivia) are removed, so that performance reflects video understanding rather than knowledge retrieval.
- **Grammar and phrasing.** Each question is checked for spelling, subject and verb agreement, punctuation, and natural English phrasing.
- **Timestep alignment.** The frames at and before the cited timestamp must contain the visual evidence required to answer the question. Items whose answer is only visible after the cited timestep are corrected to the appropriate frame range.
- **Answer correctness.** The reviewer answers the question independently from the video before consulting the proposed answer, and compares the two. Disagreements are resolved by rewatching the window together; persistent disagreements escalate to a third adjudicator.
- **Question-type matching.** The reviewer confirms that the question genuinely requires the reasoning targeted by its declared type (*contextual reasoning* for multi-step inference over a window, *instant reasoning* for single-frame recognition), and reclassifies items that are mismatched to their type.
- **Stylistic consistency.** Answers are checked against an in-house style guide: declarative sentences, present tense, third-person reference to the performer, no hedging language; numeric answers use digits, named props use the spelling agreed upon in the keyword vocabulary, and card suits and ranks use a fixed canonical form.

E Dataset Generation Pipeline

This section in the appendix details the end-to-end pipeline by which the dataset was constructed, from raw video sourcing to the final question–answer records. We utilize the VQS and QAQS outlined in (Section D) for dataset generation to maintain consistent high quality throughout the dataset. This process begins with sourcing the videos in Section E.1, and then walks through the two-pass annotation pipeline used to produce question–answer pairs (Section E.2). Then peer review that edits and validates QA pairs to make sure they meet the QAQS (Section E.3), and the final schema under which every item is stored (Section E.4). Together, these stages are designed to ensure that each benchmark item is visually grounded, unambiguously answerable, and aligned with the reasoning skill it is intended to probe.

E.1 Video Sourcing Protocol

We construct the dataset from publicly available online professional magic performances. This initial sweep yielded a pool of several hundred candidates covering a large diversity of magicians and acts.

Each video was then manually inspected end-to-end against the VQS defined in Section D, with no automated pre-filtering applied. A trained annotator watched the full clip and recorded a per-criterion verdict on the VQS dimensions. A candidate was retained only if it satisfied every criterion; failure on any single dimension was sufficient grounds for exclusion. The most common rejection causes were reliance on verbal or audio-cued reveals (failing visual self-sufficiency), aggressive editing with cuts or slow-motion replays (failing visual fidelity), and short single-effect clips lacking multiple distinct surprises (failing contextual density). Borderline cases were escalated to a second reviewer. All reviews were conducted by watching the video clips without any audio to faithfully verify if the underlying acts were understandable without audio.

E.2 Question Generation Protocol

First pass: video scan. An annotator conducts an audio-free scan through the video to fully watch the events and get a contextual understanding of the video. The annotator does not stop the video

during this phase and simply writes down a log of frames where the questions can be generated. The frames are at times in the video when non-trivial changes of object state occur (an object disappearing, materializing, transforming, or relocating). This yields a dense list of candidate moments at frame-level resolution, capturing the trick's branching structure exhaustively before any question is committed to writing. The annotator is allowed to restart the video once it has ended if any important event was missed in the log.

Second pass: question authoring. The annotator at this point has seen the full video and has built a full log of question-worthy time points in the video. The annotator then starts to draft question-and-answer pairs, choosing the question type that best probes the reasoning demanded by that moment (see Section 3.2). Each question is anchored by an explicit *timestep* (e.g., "At 0:57 of the video, . . . ") so that the model's context window is unambiguously defined at evaluation time. The annotator additionally tags the item with one or more *keywords* naming the dominant prop or technique (e.g., `Card Trick`, `Coin Trick`, `Illusions`, `Flying Trick`, `Rubik's Cube Trick`; sample vocabulary in Figure 4).

A visual example of the custom tool used to generate the questions can be seen in Figure 8.

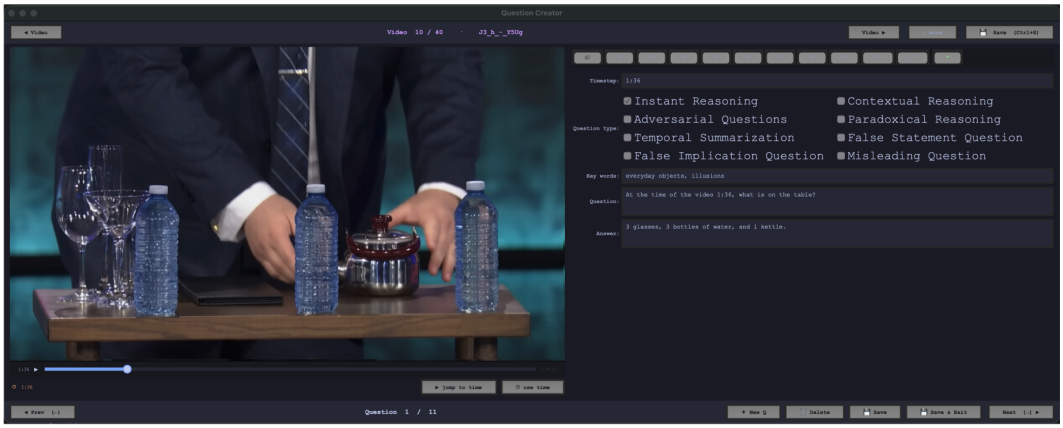


Figure 8: Sample question generation with generation tool.

E.3 Peer Review Protocol

At this stage, a group of video collectors and annotators obtained magic videos, eliminated those that did not meet the VQS, and then generated a series of Question-Answer(QA) pairs. All questions for a given video were generated by the same annotator. To validate the quality of the data, a peer review is conducted to make sure all questions meet the QAQS.

A record is admitted into the final benchmark only after passing every item on the checklist. Disagreements unresolved by the original annotator and the reviewer escalate to a senior annotator whose decision is final and is logged in the dataset's audit trail. The audit trail records, for every accepted item, the identity of the annotator, the identity of the reviewer, the dimensions on which revisions were requested, and the final adjudicated state, and is released alongside the dataset to support reproducibility.

E.4 Final Record Schema

As the QA pairs are finalized and meet the required standard, the output is structured as a single JSON file keyed by source video identifier. Each video maps to a `Questions` list of question–answer records together with a `metadata` block, so that all items grounded in the same performance are colocated and can be retrieved jointly at evaluation time. Formally, the dataset has the structure

$$\texttt{\{video_id: \{"Questions": [record}_1\texttt{, ..., record}_n\texttt{], "metadata": \{...\}\}\},}$$

where each record is a tuple of the form

$$\texttt{(UID, Timestep, Question type, Key words, Question, Answer).}$$

Field semantics. The `video_id` corresponds to the YouTube identifier of the source clip and serves as the primary key for retrieving the underlying video. Each video carries a `metadata` block whose `start` field marks the timestamp at which the magic performance proper begins, allowing the evaluation protocol to skip pre-performance introductions and stage setup. However, this data is not provided to the model for question answering. The `UID` is a globally unique integer identifier for the item, assigned independently of the video grouping so that records can be referenced unambiguously across splits and analyses. The `Timestep` anchors the question to a specific frame range in the source video and defines the model's context window at evaluation time. The `Question type` is drawn from the closed taxonomy introduced in Section 3.2. The `Key words` field is the keyword of the dominant prop or technique. Finally, `Question` and `Answer` are as defined in Section E.2.

Example record. A representative entry, abbreviated for space, is shown below.


```
{
  "RSeab6QqZok": {
       "Questions": [
           {
               "UID": 2410,
               "Timestep": "0:04",
               "Question type": "Contextual Reasoning",
               "Key words": "Body Magic Trick",
               "Question": "At 0:04 of the video, what is special about this magician?",
               "Answer": "The magician has 3 legs."
           },
     ...
           {
               "UID": 2418,
               "Timestep": "3:28",
               "Question type": "Paradoxical Reasoning",
               "Key words": "Body Magic Trick",
               "Question": "At 3:28 of the video, why are the people surprised?",
               "Answer": "The people are surprised because this man's legs are detached, but
                    he's still moving back using those three legs."
           }
       ],
       "metadata": {
           "start": "0:00"
       }
   }
}
```


F Question Types

This appendix expands the question types introduced in Section 3.2. For each of the question types, we specify the target ability the question isolates, the failure modes it is designed to surface, and its position relative to adjacent categories of questions.

F.1 Instant Reasoning (IR)

Instant Reasoning questions probe the model's ability to extract a complete, query-relevant description of the visual state at a single anchored timestep t. The reasoning window is deliberately collapsed to just one frame. Although the model receives the video context as input, the answer must be grounded in evidence visible at t.

The category serves a structural role in the ability of VLMs to comprehend videos. It isolates individual frames on which Contextual and Paradoxical Reasoning depend. If a model fails IR on a given frame, any downstream reasoning anchored to objects or events related to that frame is unreliable regardless of its long-context capabilities. This type of question also evaluates temporal slippage failure, where the model answers the question from a nearby but incorrect frame. This can lead to incorrect answers as those frames contain different information.

F.2 Contextual Reasoning (CR)

Contextual Reasoning (CR) questions ask the model to answer a query at a specific timestep using evidence established earlier in the video. The queried frame is insufficient on its own. The answer may depend on the earlier identity of the object, a state tracked through occlusion, a prior view of

a surface now hidden, or a persistent fact such as which prop was chosen or which card was signed. The model must therefore maintain a cross-frame memory of the performance and selectively retrieve the subset of that memory relevant to the query, fusing the aggregated context to produce a grounded answer.

CR questions cannot be resolved from the queried frame alone. The model must connect objects and their states across time and retrieve the earlier evidence to answer. This focus on reasoning what is true distinguishes CR from Paradoxical Reasoning (PR), which additionally asks what should have happened and why the observed outcome was unexpected. CR also differs from Temporal Summarization (TS) by answering a specific query rather than summarizing events over a time window. Common failure modes include confusing similar props introduced at different times, relying on the most recent view instead of the relevant earlier evidence, and confidently answering while overlooking a fact established much earlier.

F.3 Paradoxical Reasoning (PR)

A PR question asks why an event at a particular timestep is unexpected to a human audience. To answer the question, the model must use the preceding video to infer what should happen under ordinary physical and causal constraints, identify what actually happens, and explain the discrepancy. This distinguishes PR from CR, which can be answered by accurately tracking the observed state alone. Stating either the expected or the observed outcome in isolation does not explain the surprise.

Forming that expectation requires the model to track relevant objects through occlusion and misdirection and apply principles such as object permanence, gravity, identity under transformation, and spatiotemporal continuity. These steps can fail independently. A model may track the objects correctly but never form a prediction, leaving it with an accurate description rather than an explanation. Or it may base a plausible prediction on the wrong earlier state, producing an explanation that sounds coherent but is factually incorrect.

Magic poses a further challenge that a seemingly minor action may establish an expectation long before the reveal, with misdirection filling the interval. To explain the surprise, the model must trace the reveal back to that action and identify when and how the expectation arose. PR requires the model to pay attention to details and trace the backward causal links, testing a capability that single-frame and short-horizon evaluations cannot capture.

Question framing: We ask "Why is the audience surprised?" rather than "What is surprising?" to make the audience's perspective explicit. A model trained with magic videos may recognize the method behind a trick and find the reveal entirely trivial and nothing surprising. Asking it to identify what is surprising without specifying whose expectations would therefore leave the task ambiguous. Our formulation keeps PR meaningful even for models that can decipher the illusion. To answer correctly, the model must reconstruct what the performance leads a human viewer to believe, what the viewer would expect on that basis, and why the reveal conflicts with that expectation. This is essentially the flow of cognition and requires reasoning from the audience's understanding of the preceding events, even when the model's own knowledge supports a different prediction. The ability to decipher a trick thus does not make PR trivial. The model needs to possess the cognitive process of human viewers and explain how the performance creates a contradiction for its audience.

F.4 False Statement Question (AQ.FS)

False Statement Questions (AQ.FS) test whether a model can reject a confidently asserted false premise using video evidence. Each question states a premise contradicted by specific visual context and asks the model to reason based on it. A correct response should reject the premise and ground its reasoning based on actual visual information, while an incorrect response accepts and elaborates on the false premise, often hallucinating details that make it appear to be coherent.

AQ.FSs target a failure mode in which a model accepts a false assertion despite contradictory video evidence. Multiple-choice evaluation can obscure this failure by allowing models to exploit reasoning shortcuts without evaluating the premise. The open-ended format, by contrast, exposes the entire causal chain of the responses of the model.

AQ.FSs are not a simple correctness test. The premise in an AQ.FS is not arbitrarily wrong, but chosen to be verifiable from visual context. A successful answer therefore requires the model to (i) recognize the prompt as making a false factual claim about visual content, (ii) retrieve the relevant

visual evidence, (iii) detect the conflict between the asserted premise and the retrieved evidence, and (iv) reject the premises and perform grounded reasoning.

F.5 False Implication Question (AQ.FI)

False Implication Questions (AQ.FI) introduce false content through presupposition, using the question structure to imply something ungrounded by the video without explicitly asserting it. A correct response must identify the presupposition, check it against the visual evidence, and reject the ungrounded framing. AQ.FI reveals failures in recognizing false presuppositions and responding consistently once they are identified. To make these failures observable, we design the questions to elicit explanations of which assumptions are false and how they conflict with the visual evidence.

AQ.FSs and AQ.FIs differ in "where" they introduce a false premise. AQ.FSs assert it explicitly in a declarative statement, whereas AQ.FIs imply it through the structure of the question. Both test whether a model verifies the premise against video evidence or accepts it as ground truth to reason upon. from different angle

F.6 Temporal Summarization (TS)

Temporal Summarization questions require the model to produce a compact, abstract description of the events within a specified temporal window. A correct TS response captures essential details, while suppressing both low-level details and over-generalizations. The category isolates the ability to compress a temporal interval rather than the ability to enumerate.

TS questions mainly probe two characteristic failure modes. The first is granularity misalignment, in which the model either provides a frame-by-frame narration or overgeneralizes ("the magician performs a card trick"). The second is event omission, in which the model produces a locally accurate summary but skips a critical action within the window, often the action that motivated the question. Both failure modes are difficult to assess by multiple-choice evaluation, where the candidate answers supply predefined descriptions rather than requiring the model to summarize critical events and choose its own level of abstraction.

G Grading Pipeline

This section describes the human and machine grading protocols used to evaluate VLM responses. Human grading provides the primary benchmark scores, while machine grading provides a separate evaluation for comparison with human judgments.

G.1 Grading Protocol

After collecting the model responses, we evaluated them using both human grading and a majority vote among three API-based VLM judges. We report the two sets of scores separately and compare them across all 14 models and six question categories.

Grading the VLM Answers The two grading protocols are described below:

- **Automated Machine Grading:** We use an LLM-as-a-judge approach [41, 43] with three graders: `qwen3.7-plus-2026-05-26`, `gemini-3.5-flash-lite`, and `gpt-5.6-luna`. All graders are accessed through their official APIs with temperature 0.0 and seed 42. Each grader independently evaluates every model response using the corresponding video frames, question, ground-truth human answer, and question type, following the rubric in Figure 10. Each grader assigns a binary score of 0 (Fail) or 1 (Pass) to the model answer. The final machine grade is determined by majority voting. A response passes if at least two of the three graders assign a score of 1. Complete machine-grading results are reported in Appendix H.
- **Human Grading:** Human evaluators manually grade responses from all 14 models to obtain the primary benchmark scores. To support consistent grading across models, the workload was organized by question subsets rather than by model. Each evaluator was assigned a set of questions and graded every response of all the models to that same set. The custom grading interface used by human evaluators is shown in Figure 9.

> You are a strict grading assistant. Your sole task is to compare each submitted answer against its ground-truth answer and assign a binary score.

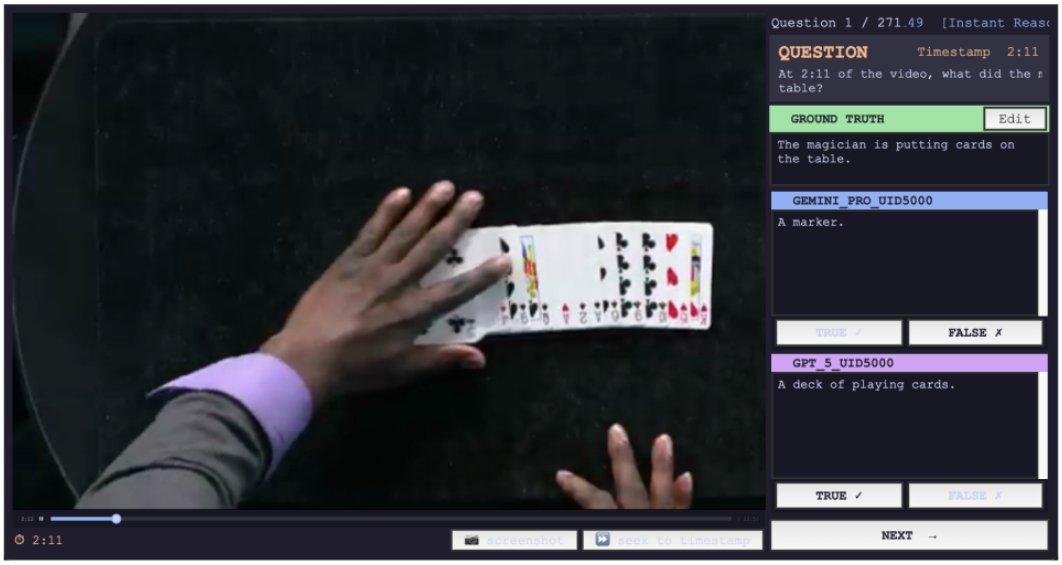


Figure 9: Custom interface used by human evaluators to grade VLM responses.

Inputs
For each grading request, you will receive:

1. Video frames.
2. A structured data object containing the questions, ground-truth answers, submitted answers, and question types associated with the video.

The input follows this conceptual structure:

```
{
    video_id: {
        uid: {
            Question: question,
            GT_Answer: ground_truth_answer,
            Submitted_Answer: submitted_answer,
            Question_type: question_type
        }
    }
}
```

The top-level key is the video ID and identifies the corresponding provided video.
Each inner key is a question UID. Preserve all video IDs and UIDs exactly in your output.
Grade every UID independently.
Inspect the provided video whenever necessary to understand the question, identify visual events, or resolve ambiguous references such as pronouns, object identities, actions, timing, and spatial relationships.

Rule 1: Ground Truth Is the Absolute Authority
The `GT_Answer` must always be treated as completely correct.
Do not override, dispute, reinterpret, or correct the ground-truth answer based on your own interpretation of the question or video.
Use the video only to understand the context and determine whether the submitted answer matches the ground truth.
If your interpretation of the video conflicts with the ground-truth answer, always follow the ground-truth answer.

Rule 2: Binary Scoring
Assign one of the following scores:

- 1 means Pass. The submitted answer captures all key facts required by the ground-truth answer at the same level of specificity. Equivalent wording is acceptable.
- 0 means Fail. The submitted answer omits a required key fact, is insufficiently specific, contains an incorrect claim, or otherwise fails to fully satisfy the ground-truth answer.

An answer that is generally correct but omits a specific detail required by the ground truth must receive 0.

Rule 3: Strict Specificity
The submitted answer must match the level of specificity of the ground-truth answer.

A broadly true statement is not sufficient when the ground truth provides a more specific answer.

Example 1:
- Question: Which hand is the ball in?
- Ground truth: The ball is in the man's right hand.
- Submitted answer: The ball is in his hand.
- Score: 0
- Reason: The submitted answer does not identify the right hand.

Example 2:
- Question: How many Kings are on the table?
- Ground truth: There are four Kings on the table.
- Submitted answer: There are some cards on the table.
- Score: 0
- Reason: The submitted answer omits both the number and the identity of the cards.

Example 3:
- Question: Why are the people surprised?
- Ground truth: The tricks were predicted and written on the board before the act started.
- Submitted answer: The magician performed a magic trick with a board, which surprised the people.
- Score: 0
- Reason: The submitted answer does not explain that the tricks were predicted and written down before the performance.

Rule 4: Question-Type Requirements
Apply the rule corresponding exactly to the value of `Question_type`.

Adversarial, False Implication
Assign 1 only when the submitted answer provides both:
1. The correct yes-or-no judgment.
2. A sufficiently specific and correct explanation.

If either component is missing, vague, or incorrect, assign 0.

Adversarial, False Statement
Assign 1 only when the submitted answer both:
1. Clearly rejects or corrects the false premise.
2. Provides the correct replacement fact.

If either component is missing, vague, or incorrect, assign 0.

Temporal Summarization
Assign 1 only when the submitted answer gives a correct and complete summary of the period specified in the question.
Assign 0 when the summary is incomplete, vague, temporally incorrect, or omits a required event.

Paradoxical Reasoning
Assign 1 only when the submitted answer gives specific and correct details explaining what is surprising, paradoxical, or unexpected.
A generic statement that something was surprising or magical is insufficient and must receive 0.

Instant Reasoning
Assign 1 only when the submitted answer correctly identifies the content, object, person, action, state, or event described by the ground truth.
A wrong identification or materially different meaning must receive 0.

Contextual Reasoning
Assign 1 only when the submitted answer correctly identifies the content, object, person, action, state, relationship, or event described by the ground truth using the relevant video context.
A wrong identification or materially different meaning must receive 0.

Rule 5: Additional Grading Rules
- Do not penalize differences in wording, grammar, sentence structure, or vocabulary when the meaning is equivalent.
- Do not require exact string matching.
- Do not penalize the omission of minor, nonessential descriptors, such as omitting wooden from wooden table, when the core object and action remain correct.
- Do not accept vague answers that avoid making an incorrect claim but fail to answer the question specifically.
- Assign 0 if the submitted answer includes clearly false, contradictory, or hallucinated information, even when the rest of the answer is correct.
- Assign 0 if an added claim materially changes or contradicts the correct answer.
- Do not award partial credit.

- Do not use information from one UID to grade another UID.
- Judge only whether the submitted answer satisfies the question and ground-truth answer for that UID.

Output Requirements
Return exactly one syntactically valid JSON object.
Do not include Markdown code fences, introductory text, concluding text, analysis, or comments outside the JSON object.
Provide one entry for every input video ID and every UID.
For each UID:

- Include a field named `reason` containing exactly one concise sentence explaining why the answer passed or failed.
- Include a field named `score` containing the integer 0 or 1.
- The reason must identify the decisive matched, missing, incorrect, vague, or hallucinated detail.
- Do not provide lengthy reasoning or hidden chain-of-thought.

The output must follow this conceptual structure:

```
{
    video_id: {
        uid: {
            reason: one concise sentence explaining the decision,
            score: 0 or 1
        }
    }
}
```

The actual response must use standard valid JSON syntax, including the quotation marks required by JSON.
Additional output constraints:

- Do not include trailing commas.
- Do not omit any video ID or UID.
- Do not add video IDs or UIDs that were not present in the input.
- Preserve the original video IDs and UIDs exactly.
- The score must be an integer rather than a string.

The video frames and corresponding grading data will be provided immediately after this instruction.

Figure 10: Prompt used by our LLM-based grading assistant. The grader receives video frames and the corresponding questions, ground-truth answers, submitted answers, and question types. It independently assigns each answer a binary score and returns a JSON object with one concise reason and score per question UID.

H Machine Grading Results

Category and overall accuracy are the percentages of responses that pass within each category and across the full dataset, respectively. Human grading remains the primary benchmark evaluation; Table 3 reports the differences between human and majority-vote machine scores for all 14 models.

NeurIPS Paper Checklist

1. **Claims**
   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?
   Answer: [Yes]
   Justification: The paper introduces *The Prestige* as an open-ended benchmark for cognitive visual reasoning based on professional magic performances, evaluations of open-source and proprietary VLMs, human/model performance comparisons, an error analysis and an ablation study showing that multiple-choice formats inflate apparent performance.
   Guidelines:
   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.
2. **Limitations**
   Question: Does the paper discuss the limitations of the work performed by the authors?
   Answer: [Yes]
   Justification: The paper discusses limitations related to reliance on YouTube as an external video source, possible future video unavailability, and the difficulty of fully aligning automatic grading with human judgment for open-ended answers.
   Guidelines:
   - The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
   - The authors are encouraged to create a separate "Limitations" section in their paper.
   - The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.

Model	**Size**	**IR**	**CR**	**PR**	**AQ.FI**	**AQ.FS**	**TS**	**Overall**
Open-source models								
Qwen3VL-Thinking [2]	8B	32.20	30.88	3.34	20.60	4.35	9.82	18.24
Qwen3VL-Instruct [2]	8B	37.56	34.43	8.29	14.59	10.87	16.07	21.68
InternVideo2.5-Chat [29]	8B	23.41	23.25	1.20	3.00	1.09	1.79	11.48
InternVL3.5 [28]	8B	17.56	23.63	1.20	15.45	6.52	1.79	12.63
MiMo-VL-SFT-2508 [36]	7B	14.15	19.31	0.80	14.59	6.52	4.46	10.66
MOSS-VL-Instruct [26]	11B	32.20	30.50	7.22	24.89	1.09	11.61	19.84
Qwen3.6 [22]	35A3B	44.88	43.20	19.12	34.33	15.22	25.89	32.06
Cambrian-S [33]	7B	18.05	19.19	1.34	12.02	0.00	4.46	10.61
Keye-VL-1.5 [32]	8B	24.88	24.90	3.34	20.60	7.61	8.04	15.43
Kimi-K2.6 [20]	1.1TA32B	52.20	45.24	24.33	33.91	26.09	27.68	35.78
Proprietary models								
GPT-5.2 [21]	UKN	47.80	42.95	27.67	15.02	8.70	39.29	33.53
Gemini3.1-Flash-Lite [13]	UKN	47.32	46.25	29.28	37.77	17.39	35.71	37.85
Qwen3.6-plus [23]	UKN	43.90	48.03	26.07	29.18	14.13	34.82	35.97
Gemini3.1-Pro [12]	UKN	50.24	48.41	37.70	41.20	56.52	34.82	43.78

Table 4: Model accuracy by question type across 14 models over the full dataset. All model results are scored by machine evaluation.

- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. **Theory assumptions and proofs**
   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?
   Answer: [N/A]
   Justification: the paper does not include theoretical results.
   Guidelines:
   - The answer [N/A] means that the paper does not include theoretical results.
   - All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
   - All assumptions should be clearly stated or referenced in the statement of any theorems.
   - The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
   - Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
   - Theorems and Lemmas that the proof relies upon should be properly referenced.
4. **Experimental result reproducibility**
   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?
   Answer: [Yes]
   Justification: The paper provides the evaluated model set, video-processing setup, 1 FPS sampling constraint, 4-minute context window, human baseline protocol, machine grading protocol, and evaluation details, which give a clear path for reproducing or verifying the results.
   Guidelines:
   - The answer [N/A] means that the paper does not include experiments.
   - If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
   - If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
   - Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general. releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in

the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.

- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
  (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
  (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
  (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
  (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. **Open access to data and code**
   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?
   Answer: [Yes]
   Justification: The paper describes the benchmark data format, release structure, audit trail, and access strategy. The dataset is documented as a JSON-style benchmark with source video identifiers, questions, answers, timestamps, question types, and metadata, and the audit trail is released alongside the dataset to support reproducibility. The code to evaluate machine performance is uploaded to Anonymous Git for inspection.
   Guidelines:
   - The answer [N/A] means that paper does not include experiments requiring code.
   - Please see the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
   - While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
   - The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
   - The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
   - The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
   - At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
   - Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. **Experimental setting/details**
   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?
   Answer: [Yes]
   Justification: The paper specifies the evaluated model set, the video-processing setting, API/cluster usage, human baseline construction, machine grading, human verification, and the multiple-choice ablation setup. Since the paper does not train new models, optimizer or training hyperparameter details are not applicable.
   Guidelines:
   - The answer [N/A] means that the paper does not include experiments.

- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment statistical significance**
   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?
   Answer: [Yes]
   Justification: The paper reports aggregate benchmark accuracies and provides mean and standard deviation for human–machine grading discrepancies across manually graded models. It also explicitly discusses grading variability and category-level differences, which is appropriate for an empirical benchmark without stochastic model training.
   Guidelines:
   - The answer [N/A] means that the paper does not include experiments.
   - The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
   - The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
   - The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
   - The assumptions made should be given (e.g., Normally distributed errors).
   - It should be clear whether the error bar is the standard deviation or the standard error of the mean.
   - It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
   - For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
   - If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. **Experiments compute resources**
   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?
   Answer: [Yes]
   Justification: The paper states that open-source models were run on A100 GPUs, while Qwen-family models, Kimi-K2.6, and closed-source models were accessed through official API portals. This identifies the main compute and access resources used for the experiments.
   Guidelines:
   - The answer [N/A] means that the paper does not include experiments.
   - The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
   - The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
   - The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. **Code of ethics**
   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?
   Answer: [Yes]
   Justification: The benchmark construction includes ethical filtering at the video-selection stage, excluding violent, unethical, inappropriate, low-quality, or visually insufficient content. The dataset also avoids redistributing source videos directly and instead provides links while respecting creator permissions.

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. **Broader impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

    Answer: [Yes]

    Justification: The paper discusses the broader evaluation impact of the benchmark: it identifies major weaknesses in current VLMs, argues for stricter open-ended evaluation, and highlights how multiple-choice formats can inflate perceived reasoning ability. It also discusses limitations and responsible release considerations.

    Guidelines:

    - The answer [N/A] means that there is no societal impact of the work performed.
    - If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
    - Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
    - The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
    - The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
    - If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. **Safeguards**

    Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

    Answer: [Yes]

    Justification: The paper includes safeguards in both dataset construction and release. Videos are manually inspected against video quality standards; inappropriate or unsuitable videos are excluded.

    Guidelines:

    - The answer [N/A] means that the paper poses no such risks.
    - Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
    - Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
    - We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. **Licenses for existing assets**

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?
Answer: [Yes]
Justification: The paper credits the models and datasets used in evaluation through citations and states that the benchmark relies on YouTube as an external source while providing video identifiers rather than redistributing source videos. It also states that the dataset is released under the CC-BY-NC-SA-4.0 license.
Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, `paperswithcode.com/datasets` has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. **New assets**
Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?
Answer: [Yes]
Justification: The paper introduces a new benchmark dataset and documents its construction process, including video sourcing, two-pass question generation, peer review, quality standards, final schema, and audit trail.
Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. **Crowdsourcing and research with human subjects**
Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?
Answer: [N/A]
Justification: The paper does not involve crowdsourcing experiments or research with human subjects. Human involvement is limited to internal dataset annotation, peer review, and benchmark grading, rather than collecting personal data or conducting a human-subjects study.
Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

    Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

    Answer: [N/A]

    Justification: The paper does not involve human-subjects research or crowdsourced participant experiments. Therefore, IRB approval or equivalent human-subjects review is not applicable.

    Guidelines:

    - The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
    - Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
    - We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
    - For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**

    Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

    Answer: [Yes]

    Justification: The paper explicitly describes LLM usage in the methodology, including LLM-as-a-judge grading and LLM-generated distractors for the multiple-choice ablation study. The machine evaluator receives the question, ground-truth answer, model response, and question type, then assigns a binary score using the stated rubric. We also use LLM to pick grammar and polish sentences.

    Guidelines:

    - The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
    - Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.